

Churn Risk & Budget-Constrained Retention Targeting

Who to target, how much to spend, and what it's worth — built on a 20,000-customer synthetic subscriber base.

20,000

active customers scored

10.6%

baseline 30-day churn rate

0.770

AUC — best model (Gradient Boosting)

3.52x

churn lift in top-risk decile

Problem Framing

- **Business objective:** protect net retained revenue over the next 30 days — not just lower the churn count. A saved customer whose offer costs more than they're worth still destroys value.
- **Prediction objective:** for every active customer, predict probability of churn in the next 30 days using account, usage, engagement and market-context features available today.
- **Decision objective:** rank by risk and, under a hard 30%-of-customers budget cap, choose who receives an offer costing 20% of monthly fee — optimizing net retained revenue, not raw churn reduction.

CONTENTS

01 Data & Key Insights

EDA, churn drivers, data quality

02 Model Performance

3 models compared, lift, drivers

03 Targeting Economics

Budget rule, cost, revenue impact

04 Scenario Comparison

Conservative vs. Aggressive, recommendation

05 Rollout, Assumptions & Next Steps

Validation plan, what to watch

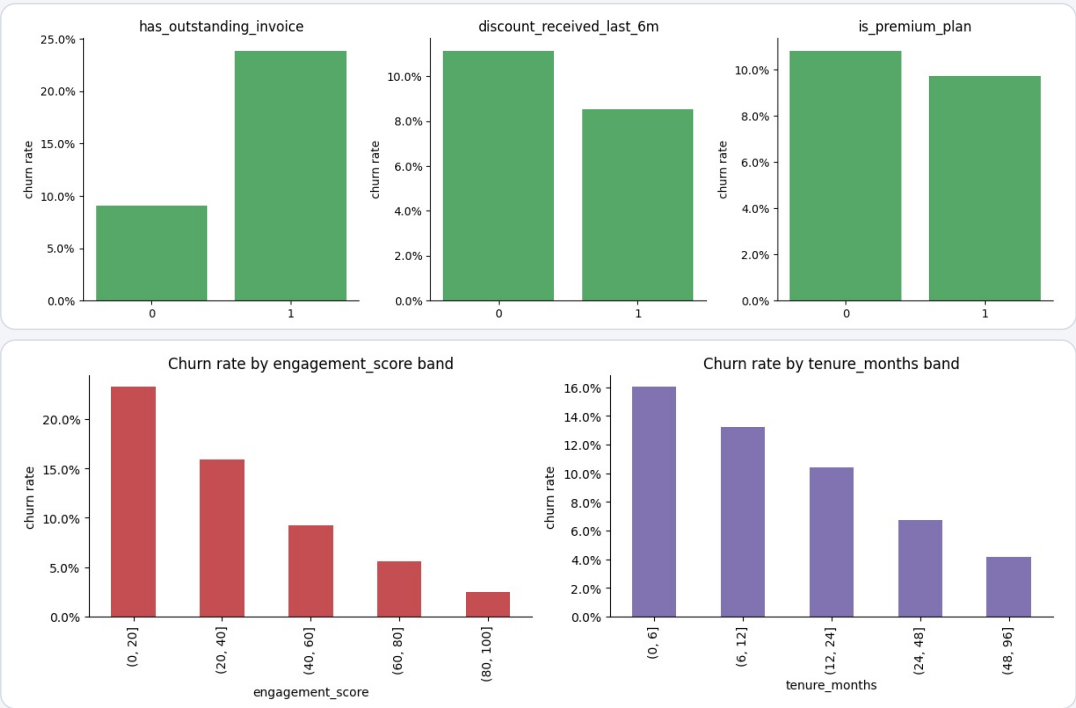
Tiers completed: **1, 2, and 3** (Foundations · Modeling & Targeting · Simulation & Experimentation). Companion notebook: *Churn_Retention_Assignment_Notebook.ipynb*.

BOTTOM LINE

A model with **0.770 AUC** and **3.52x** risk concentration is strong enough to act on — but the budget-maximal 30% targeting rule is **net-negative in month one**. Extending the lens to 3 months and narrowing the target band to the top 10% turns this into a **+\$9,355 net gain**, the best result found. Full detail on pages 3–5.

Data & Key Insights

Synthetic dataset, 20,000 customers, 13 features — generated with deliberately realistic churn drivers and data-quality noise.



What drives churn

- Low-engagement customers (score ≤ 20) churn at **23.6%** vs. **2.5%** for highly engaged customers (score ≥ 80) — the single strongest driver.
- Customers with an outstanding invoice churn at **23.9%** vs. **9.1%** without one — billing friction is a sharp, actionable signal.
- New customers (tenure ≤ 6 mo) churn at **16.1%** vs. **4.4%** for tenure ≥ 48 mo — classic early-lifecycle risk.
- Premium-plan customers churn slightly less (**9.7%**) than standard-plan customers (**10.8%**) — plan tier is mildly protective.
- Churn is fairly uniform across region (10.1%–11.2%), so region is a minor driver relative to engagement and billing.

Data quality & risk notes

- **Missingness isn't random:** ~2.0% of engagement_score and ~1.5% of total_usage_last_30d are missing — likely correlated with disengaging customers, so it's treated as signal, not noise.
- **147 billing outliers** flagged ($>3 \times \text{IQR}$ on monthly_fee) — a mix of genuine high-ARPU accounts and likely data-entry glitches; capped rather than dropped.
- **8 duplicate customer_id rows** and no post-outcome leakage features detected; all features are pre-outcome, as-of-today signals.

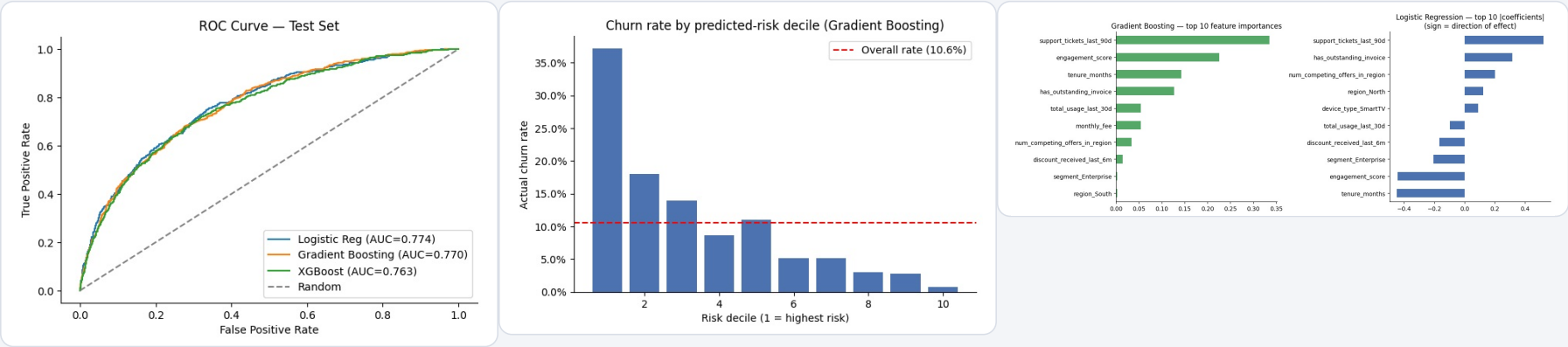
20,008 → 20,000
rows generated → clean rows after dedup

9 + 3
numeric + categorical features used

99th pct.
cap applied to monthly_fee before modeling

Model Performance

Three candidate models, stratified 75/25 train/test split (15,000 / 5,000 customers, 10.6% churn in both).



MODEL	AUC (TEST SET)	ROLE
Logistic Regression	0.774	Interpretable baseline
Gradient Boosting	0.770 ★ selected	Best AUC among boosted models — used for scoring & targeting
XGBoost	0.763	Regularized alternative, close 3rd

Reading the result: all three models land in a tight 0.76–0.77 AUC band — a stable, defensible signal rather than one model dominating by luck. Gradient Boosting is auto-selected as the production candidate; the notebook is wired to pick whichever boosted model wins, so this choice re-validates automatically on re-runs. The top risk decile churns at **3.52x** the base rate, meaning the model concentrates risk exactly where a targeting budget should be spent.

support_tickets_last_90d

#1 driver in both Gradient Boosting and Logistic Regression

tenure & engagement

strongest protective (risk-lowering) factors in both models

0.011

AUC spread across all 3 models — a stable, non-fragile signal

Budget-Constrained Targeting Strategy

Rule: score all 20,000 customers, rank by churn probability, target the top 30% (the maximum budget allows).

METRIC	VALUE
Targeting rule	Top 30% by risk score (score ≥ 0.109)
Customers targeted	6,000 (30.0% of base)
Expected churners among targeted (model)	1,375
Assumed retention uplift	35% (stated assumption)
Estimated customers saved	481
Total offer cost (20% of fee × targeted)	\$28,574
Est. 1-month retained revenue	\$11,462
Net retained revenue impact (1 month)	-\$17,112

Assumption module: ASSUMPTIONS = {"max_target_pct": 0.30, "offer_cost_pct": 0.20, "retention_uplift": 0.35} — every figure above recalculates from these three inputs.

\$4.76

avg. offer cost per targeted customer

22.9%

model-expected churn rate inside the targeted band

2.16x

band risk concentration vs. 10.6% base rate

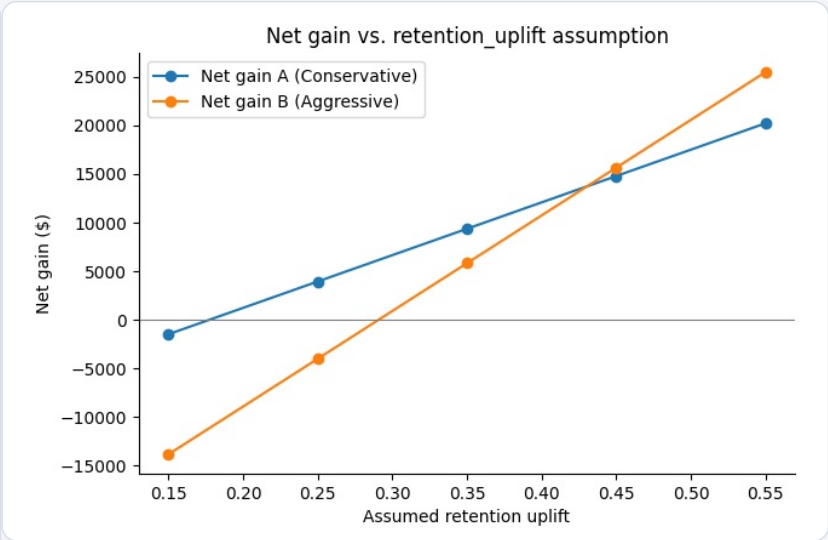
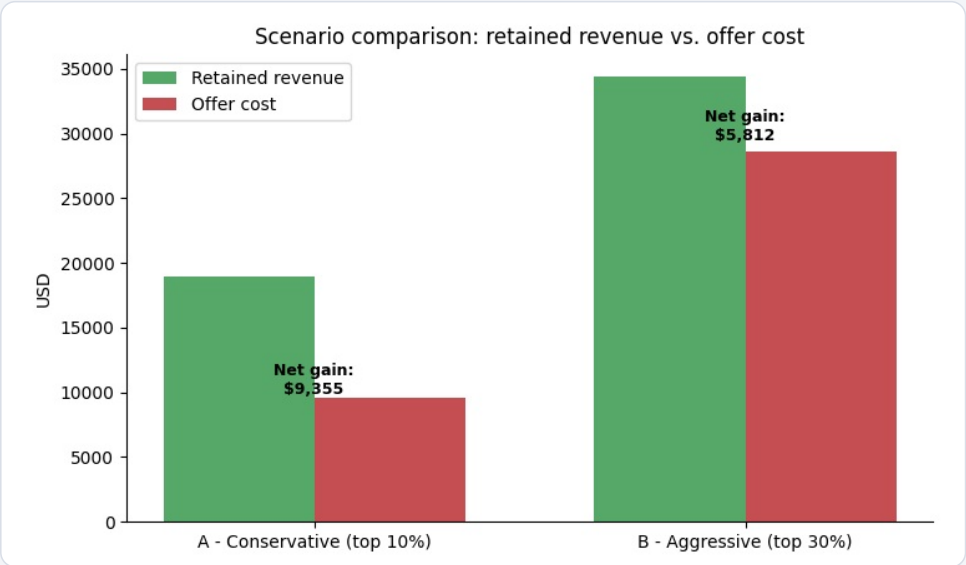
The 1-month view is intentionally the "before" picture.

Spending the **full** 30% budget on a pure risk-rank rule costs \$28,574 up front but only recovers \$11,462 within the first month — a **net loss of \$17,112** if we stop measuring there. That's not a flaw in the model; it's what a single-month lens misses: the value of a saved subscriber compounds over their retained tenure, not in month one alone.

What this means for the decision: net impact should be judged over a realistic revenue horizon, and the targeting *band width* matters as much as the model's accuracy. Both are tested head-to-head on the next page over a 3-month horizon.

Scenario Comparison — 3-Month Horizon

Same model, same cost/uplift assumptions — only the size of the targeted risk band changes.



SCENARIO	TARGETED	SAVED	REVENUE (3MO)	COST	NET GAIN
A — Conservative (top 10%)	2,000	263	\$18,978	\$9,623	\$9,355
B — Aggressive (top 30%)	6,000	481	\$34,386	\$28,574	\$5,812

CHURN & RETENTION TARGETING

RECOMMENDATION

Scenario A (Conservative, top 10%) wins on net gain — \$9,355 vs. \$5,812 for the full 30% budget — despite saving fewer customers in absolute terms (263 vs. 481). Precision beats volume: the marginal customers added by widening the band (deciles 4–10) are priced-in at lower risk and lower expected payoff per dollar spent.

Sensitivity (left): the ranking holds across a wide range of retention-uplift assumptions (15%–55%) — not a knife-edge result tied to the 35% assumption.

Rollout Plan, Assumptions & Next Steps

How we'd validate this before committing real budget — and what would change the recommendation.

Validation plan

- **Randomized control/treatment test first** — hold out ~20% of the eligible/targeted group as a scored-but-not-offered control to get an unbiased read on true retention uplift, instead of relying on the 35% assumption.
- **Primary KPI: net retained revenue** (retained revenue — offer cost), matching the business objective — not churn-rate reduction alone.
- **Secondary KPIs:** churn-rate delta, offer redemption/ROI, and a guardrail metric (support tickets/complaints) to catch a "discount trains bad behavior" effect.
- **Start narrow, expand only if it holds:** launch with the Conservative (top 10%) band — lower exposure while the uplift assumption is validated — before considering the full 30% budget.
- **Duration:** at least one full 30-day churn window, ideally 2–3 cycles, sized to detect the assumed uplift at conventional significance.
- **Decision gate:** expand budget only if treatment's net retained revenue is positive and statistically distinguishable from control, with no guardrail regression.

Key assumptions (all adjustable)

Max target share of base	30%
Offer cost	20% of monthly fee
Retention uplift from offer	35% (assumed)
Revenue horizon (Tier 3)	3 months

Next steps

- Replace the assumed 35% uplift with a measured value from the control/treatment test.
- Move to a monitored, retrainable pipeline with quarterly drift checks (e.g., competitive pressure by region).
- Extend the revenue model for partial-month churners and multi-offer customers.

CLOSING SUMMARY

The model discriminates churn risk reliably (0.770 AUC, 3.52× top-decile lift), and the economics point to a clear starting play: target the **top 10% highest-risk customers** rather than the full 30% budget, expect roughly **\$9,355 in net retained revenue** over 3 months, and validate the 35% retention-uplift assumption with a real control/treatment test before expanding spend.